

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____ GUO Qirui _____

Student ID: _____ 22099436d _____

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? No

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

All data sending to 192.168.2.0/24 from R3 will be routed through R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65=64+1

64: T1 serial link (1.544Mb/s)

1: R2 Gigabit 0/0 LAN link

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

% OSPF: Reference bandwidth is changed.

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

To obtain a higher accuracy cost for faster links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 + R3 G0/0, 781+1=782

192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1, 781+64=845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562

Each serial link has a cost of 781, and the route to 192.168.23.0/24 has two serial links.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the least estimated total cost route. The cost of the route through R2 is less than the route direct from R1 to R3

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID (RID) serves as the unique identifier for each router in the OSPF domain.

Each router must have a unique RID. If two routers have the same RID, OSPF will malfunction. By explicitly assigning the RID (via router-id command or a loopback interface), the ID remains stable even if physical interfaces go up/down. On multi-access networks, the router with the highest RID is elected as DR. Uncontrolled RID assignment can lead to unpredictable DR elections, potentially selecting a low-performance router as DR. If the RID changes dynamically, all OSPF adjacencies are reset and the SPF algorithm re-runs, causing network instability. Explicit RID assignment prevents this.

2. Why is the DR/BDR election process not a concern in this lab?

The serial link used in this lab are point to point links. The DR/BDR election process is not a concern in this lab because the OSPF network topology consists of only point-to-point links or a simple topology where no multi-access networks require DR/BDR election.

3. Why would you want to set an OSPF interface to passive?

It can free up bandwidth by eliminating unnecessary OSPF information. And to prevent OSPF from sending hello packets and forming adjacencies on an interface while still advertising the network.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of areas in OSPF enables hierarchical network design, reducing routing overhead and improving scalability.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used.

When a router receives routes to the same destination from different routing protocols, it selects the route with the lowest Administrative Distance. Metrics are only compared within the same routing protocol.